

Emanuele D'Ossualdo

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Current position

Apr 2024–present **Tenure-Track Professor**, Formal Methods for Software Engineering,
University of Konstanz, Germany.

Experience

- Sep 2020–Mar 2024** **Postdoctoral Researcher**, Foundations of Programming group,
Max Planck Institute for Software Systems, Saarbrücken, Germany.
Topic Concurrent Separation Logics, Hyperproperties, Non-Volatile Memory Models,
Refinement (with Prof. Derek Dreyer).
- Sep 2018–Aug 2020** **Marie Curie Research Fellow**, Computing Department, *Imperial College London, UK.*
Topic Compositional verification and specification for progress and security properties of
concurrent software, integrating separation logics, automata theory and process algebra.
Funding Two years EU funded fellowship (H2020-MSCA-IF-2017 795218).
- Apr 2017–Sep 2018** **Research Associate**, *Imperial College London, UK.*
Topic Concurrent Separation Logic (with Prof. Philippa Gardner).
- May 2015–Apr 2017** **Postdoctoral Researcher**, *Concurrency Theory Group, TU Kaiserslautern, Germany.*
Topic Logics and Automata for Infinite State Model Checking (with Prof. Roland Meyer).

Education

- 2010–2015** **PhD in Computer Science**, *University of Oxford, Merton College, UK.*
Thesis Verification of Message Passing Concurrent Systems (supervisor: Luke Ong).
Awards [Winner of the 2016 BCS/CPHC Distinguished Dissertation award.](#)
- 2007–2010** **M.Sc. in Computer Science**, *University of Udine, Italy, 110/110 cum laude.*
Dissertation on static analysis of Bigraphs by Abstract Interpretation.
- 2004–2007** **B.Sc. in Computer Science**, *University of Udine, Italy, 110/110 cum laude.*
Dissertation on Monads and Arrows in Haskell.

Awards & Fellowships

- 2018** **Marie Skłodowska-Curie Individual Fellowship**, *EU Horizon 2020.*
Grant Title Verification and Specification through Progress Abstractions (VeSPA).
Budget € 195.454,80 for 2 years (grant number 795218).
- 2016** **Winner of the Distinguished Dissertation award**, *BCS/CPHC, UK.*
Best British PhD dissertation in Computer Science selected by the Council of Professors
and Heads of Computing, and the BCS Academy of Computing.
- 2010–2013** **Scatcherd European Scholarship**, *University of Oxford, UK.*
University-wide fully-funded PhD scholarship.
- 2004–2010** **Scuola Superiore Student Fellowship**, *University of Udine, Italy.*
University-wide fully-funded 5 years scholarship for excellent students. Members are
annually reviewed and required to attend extra courses. See scuolasuperiore.uniud.it.

Publications

- POPL'26 *Y. Matsushita, K. Hirata, R. Wakizaka, E. D'Oswaldo. RapunSL: Untangling Quantum Computing with Separation, Linear Combination and Mixing.*
Proc. ACM Program. Lang. 10, POPL. 2026.
- POPL'25 *J. Bao, E. D'Oswaldo, A. Farzan. Bluebell: An Alliance of Relational Lifting and Independence For Probabilistic Reasoning.*
Proc. ACM Program. Lang. 9, POPL. 2025.
- ESOP'25 *F. Stutz, E. D'Oswaldo. An Automata-theoretic Basis for Specification and Type Checking of Multiparty Protocols.*
Programming Languages and Systems, ESOP. 2025.
- POPL'23 *E. D'Oswaldo, A. Raad and V. Vafeiadis. The Path to Durable Linearizability.*
Proc. ACM Program. Lang. 7, POPL. 2023.
- POPL'23 *M. Sammler, S. Spies, Y. Song, E. D'Oswaldo, R. Krebbers, D. Garg, and D. Dreyer. DimSum: A Decentralized Approach to Multi-language Semantics and Verification.*
Proc. ACM Program. Lang. 7, POPL. 2023.
- [Distinguished Paper Award at POPL'23](#)
- OOPSLA'22 *E. D'Oswaldo, A. Farzan and D. Dreyer. Proving Hypersafety Compositionally.*
Proc. ACM Program. Lang. 6, OOPSLA2. 2022.
- OOPSLA'22 *D. Frumin, E. D'Oswaldo, B. van den Heuvel, and J. A. Pérez. A Bunch of Sessions: A Propositions-as-Sessions Interpretation of Bunched Implications in Channel-Based Concurrency.* Proc. ACM Program. Lang. 6, OOPSLA2. 2022.
- TOPLAS'21 *E. D'Oswaldo, J. Sutherland, A. Farzan and P. Gardner. TaDA Live: Compositional Reasoning for Termination of Fine-grained Concurrent Programs.*
In ACM Transactions on Programming Languages and Systems (TOPLAS). ACM. 2021.
- Presented at POPL'22 (Journal-first submission).
- CONCUR'20 *E. D'Oswaldo, F. Stutz. Decidable Inductive Invariants for Verification of Cryptographic Protocols with Unbounded Sessions.*
In Proc. of Concurrency Theory. LIPIcs. 2020.
- CSF'17 *E. D'Oswaldo, L. Ong and A. Tiu. Deciding Secrecy of Security Protocols for an Unbounded Number of Sessions: The Case of Depth-bounded Processes.*
In Proc. of Computer Security Foundations. IEEE Computer Society. 2017.
- LICS'16 *E. D'Oswaldo, R. Meyer and G. Zetsche. First-order Logic with Reachability for Infinite-State Systems.* In Proc. of Symposium on Logic in Computer Science. ACM. 2016.
- ESOP'16 *E. D'Oswaldo, L. Ong. On Hierarchical Communication Topologies in the π -calculus.* In Proc. of European Symposium on Programming. Vol. 9632 of LNCS. Springer. 2016.
- SAS'13 *E. D'Oswaldo, J. Kochems and L. Ong. Automatic Verification of Erlang-Style Concurrency.* In Proc. of Static Analysis. Vol. 7935 of LNCS. Springer. 2013.
- AGERE'12 *E. D'Oswaldo, J. Kochems and L. Ong. Soter: an Automatic Safety Verifier for Erlang,*
In Proceedings of the 2nd edition on Programming systems, languages and applications based on actors, agents, and decentralized control abstractions. ACM. 2012.
- Monograph *E. D'Oswaldo. Verification of Message Passing Concurrent Systems.*
BCS/CPHC Distinguished Dissertation Award Series, ISBN 978-1-78017-363-4, BCS. 2016.

Student supervision

- 2017–2022** Assistant Supervisor of **Julian Sutherland**, PhD in Computer Science, *Imperial College*.
Topic Compositional Termination Proofs of Fine-grained Concurrent Programs.
- 2019** **Felix Stutz**, MSc Computer Science, *Saarland University*, Germany.
Topic Automatic verification of cryptographic protocols through inductive invariants.
- 2019** **Ruhi Choudhury**, MEng Computing, *Imperial College London*.
- 2018** **Blaine Rogers**, MEng Joint Mathematics and Computing, *Imperial College London*.
Thesis A π -calculus Abstraction for Erlang.
– Winner of Davis Prize award (best JMC thesis).

Academic Activities

- Chair** Co-chair of the FORTE 24 Artifact Evaluation Committee.
- Program Committee** ESOP 2026, OOPSLA 2026, ICE 2024/2025, POPL 2024, CONCUR 2024, VSTTE 2024, FORTE 2024, Erlang Workshop 2018/2021/2023, EXPRESS/SOS 2019.
- Organisation** Local organiser for MFPC/CALCO 2019 in London.
- Reviewer** POPL 2025, PLDI 2018/2023/2025, LICS 2025, ICALP 2025, MFCS 2025,
Conferences OOPSLA 2020/2022/2023, ECOOP 2022/2023, CONCUR 2015/2017/2018/2020, LMCS 2020/2024, PLACES 2020, EXPRESS 2019, iFM 2019, ESOP 2019, CAV 2019, SAS 2018, ERLANG 2018/2021, FoSSaCS 2017, TACAS 2016, NETYS 2016/2021, MFCS 2012, LICS 2015, FSTTCS 2015, VMCAI 2014, DMC 2014, TAMC 2012, POPL 2012, TLCA 2011.
- Journals* TOPLAS, TCS, TOCL, Formal Aspects of Computing, Information and Computation, Information and Software Technology, Mathematical Structures in Computer Science.
- Artifact Evaluation* OOPSLA 2022/2023, ECOOP 2022/2023.
- Award Committee* POPL Student Research Competition 2021 Selection Committee.